University of Maryland at College Park

Dept. of Aerospace Engineering

ENAE 641: Linear System Dynamics

Problem Set #3

Issued: 18 Oct. 2026
Due: 27 Oct. 2026

Question 1:

The file `F16.mat` on ELMS contains a standard state-space perturbation model for the longitudinal dynamics of an F16 aircraft in level flight at 10,000 ft moving with speed 500 ft/sec. The four states of this model are: speed, angle of attack, pitch angle, and pitch rate, in that order.

a.) Determine the real block modal form for the $\boldsymbol{A}$ matrix, and show that the system is stable.

b.) Specify a physical (i.e. real-valued) initial condition which will uniquely excite the phugoid (slowly oscillating) mode. Repeat for the short period (faster oscillating) mode. Interpret each in terms of the physical variables.

c.) The two inputs to this model are throttle and elevator deviations from trim, repectively. Which mode is most strongly excited by by the throttle? Which mode is most strongly excited by the elevator? Hint: look at the corresponding modal $\boldsymbol{B}$ matrix for the transform in a.)

Question 2:

Consider a time-periodic system from Homework #2:

$$\boldsymbol{A}(t) = \begin{bmatrix} -1 + 1.5\cos^2(t) & 1 - 1.5\sin(t)\cos(t) \\ -1 - 1.5\sin(t)\cos(t) & -1 + 1.5\sin^2(t) \end{bmatrix}$$

This system is time-periodic, hence it admits a Floquet decomposition. Determine the components $(\boldsymbol{P}(t).\boldsymbol{Q}, \boldsymbol{R})$ of this decomposition.

Question 3:

For the linear time-varying system

$$\dot{\mathbf{x}}(t) = \boldsymbol{A}(t)\mathbf{x}(t) \qquad \mathbf{x}(t_0) = \mathbf{x}_0$$

if

$$\boldsymbol{A}(t) = \begin{bmatrix} 0 & t\sin t\cos\sqrt{t} \\ -t\sin t\cos\sqrt{t} & 0 \end{bmatrix}$$

and $\mathbf{x}_0^T = [-1\ 2]$, find $\|\mathbf{x}(t)\|_2$ for all $t \geq t_0$. What does this tell you about the stability properties of this system (unstable, asymptotically stable, or stable)?

Question 4:

For the F16 model in Question #1, find a constant, positive-definite matrix $\boldsymbol{P}$ so that

$$\frac{d}{dt}\left(\mathbf{x}^T(t)P\mathbf{x}(t)\right) = -\|\mathbf{x}(t)\|^2$$

(Use Matlab's `lyap` function, but read its documentation carefully!) Verify this $\boldsymbol{P}$ is positive definite.

b.) Use this $(\boldsymbol{P}, \boldsymbol{Q})$ pair to bound the exponential convergence of $\|\mathbf{x}(t)\|$ from an arbitrary $\mathbf{x}_0$.

Question 5:

a.) For the F16 model in Question #1, what are the eigenvalues of $\boldsymbol{A} + \boldsymbol{A}^T$? Do these eigenvalues allow you to conclude the kind of exponential stability result that you obtained in Question #1 and Question #4?

b.) Is there a contradition between the results of a) and that of Questions #1,4? Explain carefully in light of Lyapunov theory (HINT: What $\boldsymbol{P}$ are you impicitly using in the test of a)? Should you expect an arbitrary choice such as this to satisfy Lyapunov's requirement?)

Question 6:

For the homogeneous dynamics of a linear system, suppose that the symmetric matrices $\boldsymbol{P}(t), \boldsymbol{Q}(t)$ satisfy the Lyapunov equation, with $\boldsymbol{P}(t)$ bounded and uniformly positive definite, and $\boldsymbol{Q}(t)$ bounded. Let $\nu(t)$ be the minimal eigenvalue of $\boldsymbol{Q}(t)$, i.e.

$$\nu(t) = \min_k \lambda_k(\boldsymbol{Q}(t))$$

Determine the "strongest" property (in order: inconslusive, stable, asymptotically stable, or exponentially stable) you can conclude for the following cases below. Show your reasoning carefully, and note that some or all of these cases might not fit neatly into the categories specifically considered in class.

a.) $\nu(t) = 1/t$ b.) $\nu(t) = -ce^{-t}$ with $c > 0$.

c.) $\nu(t) = 1/t^2$ d) $\nu(t) = 2 - 3\sin(t)$.

Question 7:

Generalize your observations from Question #6. What general conditions on $\nu(t)$ would allow you to conclude isL stability? Asymptotic stability? Exponential stability? NOTE: this question is asking you to go beyond the comparatively straightforward results discussed in lecture; for example to demonstrate formally that stability can sometimes be concluded if $\nu(t) < 0$ for some or even all $t \geq t_0$ etc.